

Full fairness does not force independence of prefix coordinates

Exact obstruction to a proposed rank argument; independently reviewed

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This note concerns a possible route to a factorial lower bound for the total number of faces. It shows that full permutation fairness does not force the raw ordered-prefix coordinates to be linearly independent. It does not settle regularity of the endpoint map in the reduced Lie algebra, whose tangent vectors involve conjugation by both prefixes and their inverses.

For a finite word T , let c_i be its total number of occurrences of letter i . For a word $u = i_1 \cdots i_d$ with distinct letters and a cut t in T , define

$$F_u(t) = \frac{\#\{\text{subsequences } i_1 \cdots i_d \text{ entirely before } t\}}{c_{i_1} \cdots c_{i_d}}.$$

Thus F_u is a normalized prefix coordinate of the axis path of T . One may either use cuts between letters or interpolate each individual letter occurrence continuously; the conclusion below holds in both conventions.

Proposition 1. *For every $n \geq 2$ there is a fully permutation-fair n -letter word T and a distinguished letter c such that, for each $0 \leq d \leq n - 1$, the functions*

$$\{F_{uc} : u \text{ is any ordered list of } d \text{ distinct letters other than } c\}$$

are all identical on every cut of T . Consequently this family, which has $(n - 1)!/(n - 1 - d)!$ members, has linear span of dimension one whenever $d \geq 1$.

Proof. Take a fair word W on the $n - 1$ old letters, with old counts a_i . Choose a rational equal-weight quadrature $x_1, \dots, x_K$ for the uniform interval, exact through degree $n - 1$, and a common denominator N of its nodes. The rational-design insertion proposition in `main.tex` constructs a fair word T by concatenating N copies of W and inserting one c in gap Nx_ℓ for every node. Repeated nodes give consecutive occurrences of c . All old letters have total counts Na_i , and c has total count K .

Before a c in gap j , the restriction of the prefix to old letters is W^j . Self-concatenation preserves fairness. Hence, for every ordered list u of d distinct old letters, its normalized count in that prefix is

$$\frac{j^d \prod_{i \in u} a_i / d!}{\prod_{i \in u} Na_i} = \frac{(j/N)^d}{d!}.$$

The coordinate F_{uc} increases only at occurrences of c . At the occurrence corresponding to node x_ℓ , its increment is therefore $x_\ell^d / (Kd!)$, independently of u . All these coordinates start at zero, so at every cut

$$F_{uc}(t) = \frac{1}{Kd!} \sum_{\ell: \text{the corresponding } c \text{ lies before } t} x_\ell^d.$$

This also proves the interpolated assertion. At the final cut the common value is $1/(d + 1)!$ by quadrature exactness, so it is nonzero. $\square$

The construction works with unequal old counts as well. Thus the collapse holds across different old subsets, not just permutations of one fixed subset. In particular $F_{abc} = F_{bac}$ for distinct old letters a, b , even in fully fair families with arbitrarily many dice. A Gram matrix of these raw prefix functions is singular regardless of how high a fairness strength is assumed. A valid lower-bound argument using endpoint derivatives would need additional structure beyond independence of these prefix coordinates; the proposition does not assert that their equality produces a Lie-algebra tangent annihilator.